J. Derek Tucker

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Education

2014 **Ph.D., Statistics**, *Florida State University*, Tallahassee, FL
Dissertation: *Functional Component Analysis and Regression Using Elastic Methods*
Advisors: Anuj Srivastava and Wei Wu

2009 **M.S., Electrical Engineering**, *Colorado State University*, Fort Collins, CO
Thesis: *Coherence-Based Underwater Target Detection for Side-Scan Sonar Imagery*
Advisor: Mahmood R. Azimi-Sadjadi

2007 **B.S., Electrical Engineering**, *Colorado State University*, Fort Collins, CO
Cum Laude; Minor in Mathematics

Appointments

Academic

2020–Present **Adjunct Professor**, *Department of Statistics, University of Illinois Urbana-Champaign*, Champaign, IL
Teach a graduate topics course in functional data analysis, advise and serve on graduate committees, and take part in the life of the department.

Professional

2022–Present **Distinguished Member of the Technical Staff**, *Sandia National Laboratories*, Albuquerque, NM
Statistical Sciences Department
- Lead research programs in elastic functional data analysis, Bayesian model calibration, and scalable density-based clustering, including PI on NNSA- and OGA-funded projects.
- Statistics subject-matter expert for cybersecurity programs.
- Build and lead multidisciplinary teams that apply statistical methods to national security mission problems.
- Mentor postdoctoral researchers, graduate student interns, and junior staff across multiple departments and centers.

2017–2022 **Principal Member of the Technical Staff**, *Sandia National Laboratories*, Albuquerque, NM
- Led research in nuclear forensics using functional and shape data analysis (PI, NTNFC, $600K).
- Led research in spatio-temporal correlation of events using point process models.
- Team lead on multiple sensor exploitation and long-haul communications teams.

2014–2017 **Senior Member of the Technical Staff**, *Sandia National Laboratories*, Albuquerque, NM

- PI, Fellow-Directed LDRD project on spatio-temporal correlation of points ($755K).
- Developed registration algorithms for remote sensing imagery that quantify registration uncertainty.
- Developed methods for functional statistical process control and statistical analysis of Doppler signatures.
- Team lead on multiple sensor exploitation and long-haul communications teams.

2009–2014 **Research Engineer**, *Naval Surface Warfare Center, Panama City Division*, Panama City, FL

- Led automatic target recognition algorithm development for sonar-based target detection and classification.
- Developed alignment and statistical analysis methods for sonar signals using a new elastic metric.
- Transitioned theoretical methods to testing and eventual use in Navy systems.

2007–2009 **Research Engineer**, *Information System Technologies, Inc.*, Fort Collins, CO

Developed transient-signal detection algorithms for sensor networks, including an HMM-based sniper detection and localization system implemented on embedded and FPGA sensor hardware.

2006–2009 **Graduate Research Assistant**, *Colorado State University*, Fort Collins, CO

Developed coherence-based detection and classification algorithms for underwater targets in sonar imagery; benchmarked satellite cloud detection using MSG-SEVIRI data.

Honors and Awards

2024 Wiley–TIES Best Paper Award, *Environmetrics*, for "Elastic Functional Changepoint Detection of Climate Impacts from Localized Sources" (with D. Yarger)

2024 Sandia National Laboratories Mission Innovation Award

2019 IEEE Senior Member

2019 Sandia National Laboratories Mission Innovation Award

2017 Director of National Intelligence Team Award, IARPA SLiCE (Signal Location in Complex Environments)

2017 Sandia ERA Team Award, "Venom Spider Bite"

2016 Sandia ERA Team Award, "Mica Data Link"

2016 Mortara Award, Computing in Cardiology, for "An Index for T-Wave Pointwise Amplitude Variability Quantification" (with J. Ramirez, M. Orini, E. Pueyo, and P. Laguna)

2013–2014 Outstanding Reviewer, *IEEE Journal of Oceanic Engineering*

Publications

Name in **bold**; † denotes a student advisee.

Peer-Reviewed Journal Articles

[J1] N. Winovich, L. Moynihan, O. Abdelrahman, R. West, S. Dauphin, **J. D. Tucker**, G. Huerta, K. Potter, R. Forrest, C. Phillips, and M. Ebeida. VoroClust: Scalable clustering for remote sensing. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 19:15071–15089, 2026.

[J2] K. Rumsey, D. Francom, G. Gibson, **J. D. Tucker**, and G. Huerta. Bayesian adaptive polynomial chaos expansions. *Stat*, 15(1):e70151, 2026.

[J3] K. Goode, **J. D. Tucker**, D. Ries, and H. Hofmann. Explainable machine learning for functional data. *Journal of Data Science*, 24(1):125–145, 2026.

[J4] A. Heiskala, **J. D. Tucker**, P. Choudhary, R. Nedelec, J. Ronkainen, O. Sarala, M. Järvelin, M. J. Sillanpää, and S. Sebert. Timing based clustering of childhood BMI trajectories reveals differential maturational patterns; study in the Northern Finland Birth Cohorts 1966 and 1986. *International Journal of Obesity*, 49(5):872–880, 2025.

[J5] D. Francom, **J. D. Tucker**, J. G. Huerta, K. Shuler, and D. Ries. Elastic Bayesian model calibration. *SIAM/ASA Journal on Uncertainty Quantification*, 13(1):195–227, 2025.

[J6] J. E. Borgert, J. Hannig, **J. D. Tucker**, L. Arbeeva, A. N. Buck, Y. M. Golightly, S. P. Messier, A. E. Nelson, and J. S. Marron. Elastic shape analysis of movement data. *Journal of the American Statistical Association: Case Studies and Applications*, 2025.

[J7] A. McCombs, M. A. Ausdemore, K. Goode, J. G. Huerta, K. Shuler, **J. D. Tucker**, A. Zhang, and D. Ries. Inverse prediction of PuO_2 processing conditions using Bayesian seemingly unrelated regression with functional data. *Frontiers in Nuclear Engineering*, 3, 2024.

[J8] **J. D. Tucker** and D. Yarger. Elastic functional changepoint detection of climate impacts from localized sources. *Environmetrics*, 35(1):e2826, 2024. *2024 Wiley–TIES Best Paper Award.*

[J9] J. Upston†, D. Sulsky, **J. D. Tucker**, and Y. Guan. CIEL*Ch color map for visualization and analysis of sea ice motion. *Journal of Computational and Applied Mathematics*, 429:115126, 2023.

[J10] K. Rumsey, J. G. Huerta, and **J. D. Tucker**. A localized ensemble of approximate Gaussian processes for fast sequential emulation. *Stat*, 12(1):e576, 2023.

[J11] D. Ries, A. Zhang, **J. D. Tucker**, K. Shuler, and M. Ausdemore. A framework for inverse prediction using functional response data. *Journal of Computing and Information Science in Engineering*, 23(1):011002, 2023.

[J12] J. L. Brown, J. P. Davis, **J. D. Tucker**, J. G. Huerta, and K. Shuler. Quantifying uncertainty in analysis of shockless dynamic compression experiments on platinum. II. Bayesian model calibration. *Journal of Applied Physics*, 134(23):235902, 2023.

[J13] **J. D. Tucker**, M. T. Martinez, and J. M. Laborde. Dimensionality reduction using elastic measures. *Stat*, 12(1):e551, 2023.

[J14] T. Harris†, B. Li, and **J. D. Tucker**. Scalable multiple changepoint detection for functional data sequences. *Environmetrics*, 33(2):e2710, 2022.

[J15] M. A. Ausdemore, A. McCombs, D. Ries, A. Zhang, K. Shuler, K. Goode, **J. D. Tucker**, and J. G. Huerta. A probabilistic inverse prediction method for predicting plutonium processing conditions. *Frontiers in Nuclear Engineering*, 1, 2022.

[J16] C. King, N. Martin, and **J. D. Tucker**. Bounding uncertainty in functional data: A case study. *Quality Engineering*, 33(1):178–188, 2021.

[J17] T. Harris†, B. Li, N. J. Steiger, J. E. Smerdon, N. Narisetty, and **J. D. Tucker**. Evaluating proxy influence in assimilated paleoclimate reconstructions—testing the exchangeability of two ensembles of spatial processes. *Journal of the American Statistical Association*, 116(535):1100–1113, 2021.

[J18] T. Harris†, **J. D. Tucker**, B. Li, and L. Shand. Elastic depths for detecting shape anomalies in functional data. *Technometrics*, 63(4):466–476, 2021.

[J19] **J. D. Tucker**, L. Shand, and K. Chowdhary. Multimodal Bayesian registration of noisy functions using Hamiltonian Monte Carlo. *Computational Statistics & Data Analysis*, 163:107298, 2021.

[J20] M. Ahn, **J. D. Tucker**, W. Wu, and A. Srivastava. Regression models using shapes of functions as predictors. *Computational Statistics & Data Analysis*, 151:107017, 2020.

[J21] **J. D. Tucker**, J. R. Lewis, C. King, and S. Kurtek. A geometric approach for computing tolerance bounds for elastic functional data. *Journal of Applied Statistics*, 47(3):481–505, 2020.

[J22] Y. Guan, C. Sampson, **J. D. Tucker**, W. Chang, A. Mondal, M. Haran, and D. Sulsky. Computer model calibration based on image warping metrics: An application for sea ice deformation. *Journal of Agricultural, Biological and Environmental Statistics*, 24(3):444–463, 2019.

[J23] **J. D. Tucker**, L. Shand, and J. R. Lewis. Handling missing data in self-exciting point process models. *Spatial Statistics*, 29:160–176, 2019.

[J24] **J. D. Tucker**, J. R. Lewis, and A. Srivastava. Elastic functional principal component regression. *Statistical Analysis and Data Mining*, 12(2):101–115, 2019.

[J25] J. Ramirez, M. Orini, **J. D. Tucker**, P. Laguna, and E. Pueyo. Variability of ventricular repolarization dispersion quantified by time-warping the morphology

of the T-waves. *IEEE Transactions on Biomedical Engineering*, 64(7):1619–1630, 2017.

[J26] T. G-Michael, B. Marchand, **J. D. Tucker**, D. D. Sternlicht, T. M. Marston, and M. R. Azimi-Sadjadi. Image-based automated change detection for synthetic aperture sonar. *IEEE Journal of Oceanic Engineering*, 41(3):592–612, 2016.

[J27] D. Bryner, F. Huffer, A. Srivastava, and **J. D. Tucker**. Underwater minefield detection in clutter data using spatial point-process models. *IEEE Journal of Oceanic Engineering*, 41(3):670–681, 2016.

[J28] D. Bryner, F. Huffer, M. M. Rosenthal, **J. D. Tucker**, and A. Srivastava. Estimation of linear target-layer trajectories using cluttered point cloud data. *Computational Statistics & Data Analysis*, 102:1–22, 2016.

[J29] **J. D. Tucker**, W. Wu, and A. Srivastava. Analysis of signals under compositional noise with applications to SONAR data. *IEEE Journal of Oceanic Engineering*, 39(2):318–330, 2014.

[J30] **J. D. Tucker**, W. Wu, and A. Srivastava. Analysis of proteomics data: Phase amplitude separation using an extended Fisher–Rao metric. *Electronic Journal of Statistics*, 8(2):1724–1733, 2014.

[J31] **J. D. Tucker**, W. Wu, and A. Srivastava. Generative models for functional data using phase and amplitude separation. *Computational Statistics & Data Analysis*, 61:50–66, 2013.

[J32] **J. D. Tucker** and M. R. Azimi-Sadjadi. Coherence-based underwater target detection from multiple sonar platforms. *IEEE Journal of Oceanic Engineering*, 36(1):37–51, 2011.

Peer-Reviewed Conference Proceedings

[C33] J. Adams, B. Berman, J. Michalenko, and **J. D. Tucker**. Conformal anomaly detection for functional data with elastic distance metrics. *Proc. Symposium on Conformal and Probabilistic Prediction with Applications (COPA), Proceedings of Machine Learning Research*, 266, 2025.

[C34] C. A. Pattyn, A. W. Edstrom, A. L. Sanchez, K. Westlake, J. D. van der Laan, **J. D. Tucker**, J. Jones, K. Hagopian, J. C. Shank, L. K. Casias, and J. B. Wright. Event-based sensing for the detection of modulated signals in degraded visual environments. *Proc. SPIE 12514, Image Sensing Technologies: Materials, Devices, Systems, and Applications X*, page 1251409, April 2023.

[C35] C. Ting, N. Johnson, U. Onunkwo, and **J. D. Tucker**. Faster classification using compression analytics. *IEEE ICDM Workshop on Data Mining and Machine Learning for Cybersecurity*, October 2021.

[C36] S. Reza, N. Martin, T. Buchheit, and **J. D. Tucker**. Tolerance bound calculation for compact model calibration using functional data analysis. *Proc. 4th IEEE Electron Devices Technology and Manufacturing (EDTM) Conference*, 2020.

[C37] L. Patel, L. Shand, **J. D. Tucker**, and G. Huerta. Assessing extreme value analysis to predict rare events from the Global Terrorism Database. *Proc. Joint Statistical Meetings (JSM)*, August 2020.

[C38] N. Martin and **J. D. Tucker**. Estimation of physical tolerance bounds for functional data. *Proc. SIAM Conference on Uncertainty Quantification*, 2020.

[C39] T. Harris[†], B. Li, N. Steiger, J. Smerdon, and **J. D. Tucker**. Evaluating proxy influence in data assimilation based climate field reconstructions. *Proc. Joint Statistical Meetings (JSM)*, August 2018.

[C40] T. Harris[†], B. Li, N. Steiger, J. Smerdon, and **J. D. Tucker**. Evaluating proxy influence in data assimilation. *Proc. Climate Informatics Workshop*, September 2018.

[C41] M. K. Ahn, **J. D. Tucker**, W. Wu, and A. Srivastava. Elastic handling of predictor phase in functional regression models. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops: Differential Geometry in Computer Vision and Machine Learning (DIFF-CVML)*, June 2018.

[C42] J. Ramirez, M. Orini, **J. D. Tucker**, E. Pueyo, and P. Laguna. An index for T-wave pointwise amplitude variability quantification. *Proc. Computing in Cardiology (CinC)*, September 2016. *Mortara Award winner*.

[C43] T. G-Michael, **J. D. Tucker**, and R. R. Roberts. Statistically normalized coherent change detection for synthetic aperture sonar imagery. *Proc. SPIE*, April 2016.

[C44] **J. D. Tucker**. Functional statistical process control using elastic methods. *Proc. Joint Statistical Meetings (JSM)*, August 2016.

[C45] T. G-Michael, B. Marchand, **J. D. Tucker**, D. D. Sternlicht, and T. M. Marston. Automated change detection for synthetic aperture sonar. *Proc. SPIE*, 9072:O1–O10, April 2014.

[C46] **J. D. Tucker**, W. Wu, and A. Srivastava. Analysis of signals under compositional noise with application to SONAR data. *Proc. MTS/IEEE Oceans 2012 Conference*, pages 1–6, October 2012.

[C47] J. C. Isaacs and **J. D. Tucker**. Signal diffusion features for automatic target recognition in synthetic aperture sonar. *Proc. IEEE DSP/SPE 2011 Workshop*, pages 461–465, January 2011.

[C48] J. C. Isaacs and **J. D. Tucker**. Diffusion features for target specific recognition with synthetic aperture sonar raw signals and acoustic color. *Proc. IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 27–32, June 2011.

[C49] **J. D. Tucker** and A. Srivastava. Statistical analysis and classification of acoustic color functions. *Proc. SPIE*, 8017:O1–O10, April 2011.

[C50] **J. D. Tucker** and N. Klausner. Compressive sensing for Gauss-Gauss detection. *Proc. IEEE SMC 2011 Conference*, pages 3335–3340, October 2011.

[C51] **J. D. Tucker**, J. T. Cobb, and M. R. Azimi-Sadjadi. Generalized likelihood ratio test for finite mixture model of K-distributed random variables. *Proc. IEEE DSP/SPE 2011 Workshop*, pages 443–448, January 2011.

[C52] N. Klausner, M. R. Azimi-Sadjadi, and **J. D. Tucker**. Multi-sonar target detection using multi-channel coherence analysis. *Proc. MTS/IEEE Oceans 2010 Conference*, pages 1–7, September 2010.

[C53] T. G-Michael and **J. D. Tucker**. Canonical correlation analysis for coherent change detection in synthetic aperture sonar imagery. *Proc. IEEE SAR/SAS 2010 Conference*, 32(4):117–122, September 2010.

[C54] **J. D. Tucker** and M. R. Azimi-Sadjadi. Neyman–Pearson detection of K-distributed random variables. *Proc. SPIE*, 7664:Q1–Q12, April 2010.

[C55] N. Klausner, M. R. Azimi-Sadjadi, and **J. D. Tucker**. Underwater target detection from multi-platform sonar imagery using multi-channel coherence analysis. *Proc. IEEE SMC 2009 Conference*, pages 2728–2733, 2009.

[C56] N. Klausner, **J. D. Tucker**, and M. R. Azimi-Sadjadi. Multi-platform target detection using multi-channel coherence analysis and robustness to the effects of disparity. *Proc. MTS/IEEE Oceans 2009 Conference*, pages 1–7, October 2009.

[C57] M. Kabatek, M. R. Azimi-Sadjadi, and **J. D. Tucker**. An underwater target detection system for electro-optical imagery data. *Proc. MTS/IEEE Oceans 2009 Conference*, pages 1–8, October 2009.

[C58] M. R. Azimi-Sadjadi and **J. D. Tucker**. Target detection from dual disparate sonar platforms using canonical correlations. *Proc. SPIE*, 6953:69530J, March 2008.

[C59] **J. D. Tucker**, N. Klausner, and M. R. Azimi-Sadjadi. Target detection in M-disparate sonar platforms using multichannel hypothesis testing. *Proc. MTS/IEEE Oceans 2008 Conference*, pages 1–7, September 2008.

[C60] **J. D. Tucker**, M. R. Azimi-Sadjadi, and G. J. Dobeck. Coherent-based method for detection of underwater objects from sonar imagery. *Proc. SPIE*, 6553:65530U, April 2007.

[C61] **J. D. Tucker**, M. R. Azimi-Sadjadi, and G. J. Dobeck. Canonical coordinates for detection and classification of underwater objects from sonar imagery. *Proc. IEEE OCEANS 2007 Conference Europe*, pages 1–6, June 2007.

Submitted Manuscripts and Preprints

[P62] G. Collins, G. Huerta, **J. D. Tucker**, and K. Rumsey. A review of Bayesian surrogate modeling for high-dimensional computer simulations. *In revision, WIREs Computational Statistics*, 2026.

[P63] **J. D. Tucker**, G. Collins, G. Huerta, and J. L. Brown. Elastic multi-fidelity Bayesian model calibration. *Submitted to SIAM/ASA Journal on Uncertainty Quantification*, 2026. arXiv:2609.27174.

[P64] D. Yarger and **J. D. Tucker**. Detecting changepoints in globally-indexed functional time series. *arXiv:2308.05915 [stat.AP]*, 2023.

[P65] R. Darling, G. S. Clark, and **J. D. Tucker**. Hidden ancestor graphs: Models for detagging property graphs. *arXiv:2102.09581 [math.PR]*, 2023.

[P66] L. Patel, L. Shand, **J. D. Tucker**, and G. Huerta. Spatio-temporal extreme event modeling of terror insurgencies. *arXiv:2110.08363 [stat.AP]*, 2021.

[P67] X. Zhang, S. Kurtek, O. Chkrebtii, and **J. D. Tucker**. Elastic k-means clustering of functional data for posterior exploration, with an application to inference on acute respiratory infection dynamics. *arXiv:2011.12397 [stat.ME]*, 2020.

[P68] A. H. Foss, R. B. Lehoucq, W. Z. Stuart, **J. D. Tucker**, and J. W. Berry. A deterministic hitting-time moment approach to seed-set expansion over a graph. *arXiv:2011.09544 [cs.SI]*, 2020.

Other Publications

[O69] **J. D. Tucker** and M. R. Azimi-Sadjadi. Coherence-based underwater target detection and classification for side-scan sonar imagery. *Sea Technology Magazine*, 12:10–14, December 2008.

Software

Lead Author

fdasrvf *R, MATLAB, Julia.* Alignment, PCA, and modeling of unidimensional and multidimensional functions using the square-root velocity framework, with elastic separation of phase and amplitude. Listed in the CRAN Task View for Functional Data Analysis.
R: CRAN and github.com/jdtuck/fdasrvf_R; MATLAB: github.com/jdtuck/fdasrvf_MATLAB; Julia (ElasticFDA.jl): github.com/jdtuck/ElasticFDA.jl

fdasrsf *Python.* Python implementation of elastic functional data analysis: alignment, PCA, and modeling of functional data using the square-root velocity framework. PyPI, conda-forge, github.com/jdtuck/fdasrsf_python, documentation

stpphawkes *R.* Hawkes-process models for spatio-temporal and temporal point processes that account for missing data (*Spatial Statistics*, 2019).
CRAN, github.com/sandialabs/stpphawkes

Developer

veesa *R.* The VEESA pipeline for explainable machine learning with functional data (*Journal of Data Science*, 2026).
CRAN, github.com/sandialabs/veesa

EFBMC *MATLAB, R, Python.* Notebooks implementing elastic functional Bayesian model calibration (*SIAM/ASA JUQ*, 2025).
github.com/sandialabs/efbmc

mvBayes *R, Python, MATLAB.* Multivariate Bayesian regression that decomposes a multivariate or functional response over a user-chosen orthogonal basis and models each component with a univariate Bayesian regression.
R, Python, MATLAB (github.com/sandialabs)

impala *R, MATLAB.* Modular Bayesian model calibration with tempering and adaptive MCMC; translated from LANL's Python impala.
R, MATLAB (github.com/sandialabs)

BASS *MATLAB.* Bayesian Adaptive Spline Surfaces; translated from LANL's pyBASS.
github.com/sandialabs/bass_matlab

BayesPPR *MATLAB.* Bayesian Projection Pursuit Regression; translated from pyBayesPPR.
github.com/sandialabs/bayesppr_matlab

Grants and Contracts

2025–2026 **Density-Based Clustering at Scale**, OGA. PI.

2024–2027 **Elastic Functional Bayesian Model Calibration**, NNSA. PI.

2018–2021 **Statistical Enhancement of Nuclear Forensics**, NTNFC. PI. $600K

2018–2020 **Data Enrichment for Improved Intelligence Value and Situational Awareness**, Sandia LDRD. PI. $1,059K

2018–2019 **Statistical Uncertainty Quantification for Multivariate Physical Parameter Estimation with Multivariate Outputs**, Sandia LDRD. Co-PI. $500K

2016–2018 **Doppler Assisted Sensor Fusion**, OGA. Co-PI (with R. M. Naething). $875K

2016–2018 **Event Correlation Using Spatio-Temporal Point Processes**, Sandia Fellow-Directed LDRD. PI. $755K

2010–2012 **In-Situ Learning for CAD/CAC Sonar Imagery**, Office of Naval Research Code 32. Co-PI (with M. R. Azimi-Sadjadi and N. Wachowski). $200K

2008–2011 **Coherence-Based Target Detection for Multi-Platform Sonar Imagery**, Office of Naval Research Code 32. Co-PI (with M. R. Azimi-Sadjadi). $200K

Invited Talks

- "Elastic Functional Changepoint Detection of Climate Impacts from Localized Sources", Environmetrics TIES Webinar, October 24, 2025
- "Bayesian Model Calibration: An Elastic Approach", Colorado State University Statistics Seminar, April 28, 2025
- "Elastic Functional Data Analysis: with Applications to Functional Changepoint", Stony Brook SIAM Speaker, November 19, 2024
- "Bayesian Model Calibration: An Elastic Approach", Purdue Statistical Colloquium, February 23, 2024
- "Elastic Bayesian Model Calibration", ASA SDNS Webinar, December 04, 2023
- "Elastic Bayesian Model Calibration", FSU Shape Group Statistical Meeting, January 24, 2023
- "Elastic Functional Data Analysis", SMU Statistical Colloquium, November 4, 2022 (Organizer: Chul Moon)
- "Elastic Functional Tolerance Bounds", LANL Statistical Colloquium, December 16, 2020 (Organizer: Emily Castleton)
- "Using FDA for Neural Network Explainability", FSU Industry Engagement Symposium, October 2, 2020 (Organizer: Elizabeth Slate)
- "Elastic Functional Data Analysis", BYU Statistical Colloquium, December 4, 2019 (Organizer: William Christensen)
- "Sandia Applications of Functional Data Analysis", UIUC Statistical Colloquium, November 22, 2019 (Organizer: Trevor Harris)
- "Elastic Depth Analysis", FSU Statistical Colloquium, April 8, 2019 (Organizer: Xufeng Niu)
- "Elastic Functional Data Analysis", ANU Statistical Colloquium, July 19, 2018 (Organizer: Janice Scealy)
- "Elastic Functional Data Analysis", UNM Statistical Colloquium, February 15, 2018 (Organizer: Li Li)
- "Event Correlation using Spatio-Temporal Point Processes", SNL DSRC Colloquium, September 29, 2016 (Organizer: K. Simonson)
- "Doppler Multi-INT Detection and Tracking", NSWC-PCD Invited Lecture, October 1, 2015 (Organizer: J. T. Cobb)
- "Alignment and Analysis of Proteomics Data using Square Root Slope Function Framework", CTW: Statistics of Time Warpings and Phase Variations 2012, OSU MBI, November 7, 2012
- "Statistical Analysis and Modeling of Elastic Functions", Department of Mathematical and Statistical Sciences, University of Alberta, August 23, 2011 (Organizer: Yau Shu Wong)

Teaching

2020–Present Graduate topics course in Functional Data Analysis, Department of Statistics, University of Illinois Urbana-Champaign

2019–Present Sandia R Training

2018–Present Sandia Statistics Primer

Student Advising and Mentoring

Ph.D. Students

2026 Carole Hall, Ph.D. Mathematics, Stony Brook University. Dissertation topic: modeling of ecological patterns using shape. Now Postdoctoral Researcher, Colorado School of Mines.

2022 Joel Upston, Ph.D. Mathematics, University of New Mexico. Dissertation topic: sea ice model calibration using proper metrics.

2021 Trevor Harris, Ph.D. Statistics, University of Illinois Urbana-Champaign. Dissertation topic: data depth and functional data analysis. Now Assistant Professor, Texas A&M University.

Graduate Committees

Current Tristan Contant, Department of Statistics, Colorado State University (expected 2027)

Postdoctoral Researchers (Sandia)

2023 Drew Yarger. Now Assistant Professor of Statistics, Purdue University.

2023 Michael Weylandt. Now Assistant Professor, Paul H. Chook Department of Information Systems and Statistics, Baruch College, CUNY.

2022 Katherine Goode. Now Staff Member, Sandia National Laboratories.

Graduate Student Interns (Sandia)

2024–2026 Carole Hall, Mathematics, Stony Brook University. Now Postdoctoral Researcher, Colorado School of Mines.

2021–2022 Jimmy Hickey, Statistics, North Carolina State University (Ph.D. 2024). Now Manager, AI/Machine Learning.

2021 David Angeles, Statistics, The Ohio State University.

2020 Robin Tu, Statistics, University of Illinois Urbana-Champaign (Ph.D. 2022). Now Senior Principal Engineer.

2020 Aubrey Odom, Statistics, Brigham Young University. Now Computational Biology Scientist, Tempus AI.

2020 Tabytha Perez.

2018–2020 Trevor Harris, Statistics, University of Illinois Urbana-Champaign. Now Assistant Professor, Texas A&M University.

Professional Service

Editorial Associate Editor, *Statistics and Computing*, May 2026–Present

Journal Referee	*Annals of Applied Statistics*; *Applied Stochastic Models in Business and Industry*; *Bayesian Analysis*; *IEEE Journal of Oceanic Engineering*; *IEEE Transactions on Aerospace and Electronic Systems*; *IEEE Transactions on Artificial Intelligence*; *JASA Case Studies and Applications*; *Journal of Computational and Graphical Statistics*; *Mathematics*; *Pattern Recognition*; *Quality Engineering*; *Statistics*; *Statistics and Computing*; *Statistics & Probability Letters*; *Technometrics*
Conference Referee	International Conference on Machine Learning (ICML)
Proposal Review	Sandia Laboratory Directed Research and Development (LDRD)

Patents

- J. D. Tucker and M. T. Martinez, "Method and System for Training an Artificial Neural Network Utilizing Physics Based Knowledge," U.S. Patent No. 12,229,934, issued February 18, 2025.
- T. G-Michael, D. Sternlicht, B. Marchand, J. D. Tucker, and T. M. Marston, "Automated Change Detection for Synthetic Aperture Sonar," U.S. Patent No. 10,049,295, issued August 14, 2018.
- "Density-Based Clustering of Multidimensional Data," U.S. patent application filed 2022 (pending).

Professional Memberships

ASA	American Statistical Association, 2011–Present
SIAM	Society for Industrial and Applied Mathematics, 2020–Present
IEEE	Institute of Electrical and Electronics Engineers, 2004–Present (Senior Member since 2019)
ISEA	2020–Present
Honor Societies	Tau Beta Pi (2006); Eta Kappa Nu (2005)

Community Service

2021–Present	Board Member, East Mountain Rodeo Association, Edgewood, NM, a 501(c)(3) charity serving East Mountain communities. Responsible for finances and event planning.
2019–Present	Shooting Sports Coach and Instructor, New Mexico 4-H
2016–2024	Back Country Horsemen of New Mexico: State Chair (2018–2023), organizing state and chapter volunteer and advocacy activities; Pecos Chapter Board Member (2016–2024), leading volunteer trail maintenance in the Santa Fe and Cibola National Forests

2006–2019 Boy Scouts of America: Assistant District Commissioner, Great Southwest Council (2017–2019); Chartered Organization Representative, Edgewood, NM (2016–2017); Venture Crew Advisor, Fort Collins, CO and Chipley, FL (2006–2016)

2002–2004 Missionary, The Church of Jesus Christ of Latter-day Saints, Fort Worth, TX

Computing

Languages Python, R, Julia, MATLAB, C, C++, Mathematica

Certifications CompTIA Security+, CompTIA Linux+

Brief Biography

J. Derek Tucker is a Distinguished Member of the Technical Staff in the Statistical Sciences Department at Sandia National Laboratories and an Adjunct Professor of Statistics at the University of Illinois Urbana-Champaign. He received the B.S. (cum laude) and M.S. degrees in Electrical Engineering from Colorado State University in 2007 and 2009, respectively, and the Ph.D. in Statistics from Florida State University in 2014 under the co-advisement of Anuj Srivastava and Wei Wu. From 2009 to 2014 he was a Research Engineer at the Naval Surface Warfare Center Panama City Division, where he led the development of automatic target recognition algorithms for synthetic aperture sonar. He joined Sandia in 2014.

His research centers on elastic functional and shape data analysis and its use in uncertainty quantification, Bayesian model calibration, statistical process control, changepoint detection, and spatio-temporal modeling, with applications ranging from climate science and remote sensing to nuclear forensics and national security. He is the lead author of the open-source fdasrvf (R, MATLAB, and Julia) and fdasrsf (Python) packages for elastic functional data analysis, as well as the stpphawkes R package for spatio-temporal point processes. He has advised Ph.D. students at the University of Illinois, the University of New Mexico, and Stony Brook University, and has mentored postdoctoral researchers who have gone on to faculty positions at Purdue University and Baruch College, CUNY.

He received the 2024 Wiley–TIES Best Paper Award from *Environmetrics*, two Sandia Mission Innovation Awards, and the Director of National Intelligence Team Award for his contributions to the IARPA SLiCE program. He is an IEEE Senior Member and serves as an Associate Editor of *Statistics and Computing*.